Gamified waste management

by Saranya K G

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1. GAMIFIED WASTE MANAGEMENT AND WASTE IDENTIFICATION

Gokuladharshini M (24Z224)
Priya K (24Z258)
Sadhana S (24Z265)
Swethi J (24Z278)
Roopa S (25Z437)

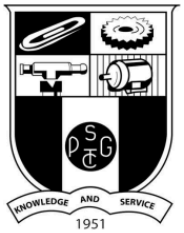
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report submitted in partial fulfillment of the requirement for the award of degree of

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(Autonomous Institution)

COIMBATORE – 641 004

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

Gamified waste management and waste identification

Bona fide record of work done by

Gokuladharshini M (24Z224)
Priya K (24Z258)
Sadhana S (24Z265)
Swethi J (24Z278)
Roopa S (25Z437)

Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: COMPUTER SCIENCE AND ENGINEERING
Of Anna University
DECEMBER 2025

……..
Dr. Saranya K G
Tutor
BE-CSE-G1

…….	…….
Dr. N.Arulanand	**Dr.G.SudhaSadasivam**
Programme Coordinator	**Head of the Department**
BE-CSE-G1	**CSE**

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CERTIFICATE

Certified that this report titled " " for the **AICTE ACTIVITY POINT PROGRAMME** project work (23Z315) is a bonafide work of **Gokuladharshini M (24Z224), Priya K (24Z258) , Sadhana S (24Z265), Swethi (24Z278), Roopa S(25Z437)** who have carried out the work for the partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering.

Place:Coimbatore
Date:

Tutor
(Dr.Saranya K G)

Department Placement Coordinator
(Dr.K SATHIYA PRIYA)

COUNTERSIGNED

Dr.G.Sudha Sadasivam
HEAD

Department of Computer Science and Engineering

PSG College of Technology,Coimbatore-641004

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ABSTRACT

In this project, we introduce a new approach to waste detection and waste management. Especially, improving user's engagement in waste management through gamification . Our main objective is to design an Innovative and technologically driven solution for stimulating active and frequent involvement with sustainable waste practices by users e.g., school children and households. We will integrate game elements including points, rewards, leader-boards, and engaging educational elements to promote active participation and to bridge the gap between stated environmental intent and habitual consistent recycling/disposal practices. The basis of the study will be a review of examples of successful gamified interventions.

We have analyzed other apps such as Junker and GreenApes and found out their successful use of features such as barcode scanning, user feedback, and educational features to promote recycling accuracy. We have also considered options to merge with new and emerging technology (IoT sensors, Artificial Intelligence based waste Classification, and cloud-based monitoring systems) to promote waste sorting and collection, participation and efficiencies.

As stated by existing behavioral theories, the Theory of Planned Behavior, and structure of emotional engagement, the project highlights the importance of motivation, awareness and user-experience in implementing environmental friendly behaviors. The new solution is a flexible and engaging system that will promote waste detection and allows community-wide waste management through extensive digital engagement.

In conclusion, the integration of mindful design and perception gained from this project will serve as an important key factor in creating a smarter, cleaner and more sustainable future by implementing gamification as a key objective in modern waste management technologies.

KEY WORDS: Gamification; Waste Management; Sustainable; Motivation; Recycling; IoT Sensors; AI-based; User Experience; Community-level; Reward

CHAPTER 1

INTRODUCTION

With respect to the growing environmental issues and the urgent need for sustainable urban development, effective waste management has become a key part of global efforts to achieve sustainability. One of the most important things to note is that as industries and cities grow, they continue to produce a large amount of waste. In fact, construction and waste production contribute to 38% of all waste generated between 2020 and 2021. Global waste production is expected to reach 3.4 billion tonnes by 2050, and new and creative techniques are essential to properly identify, separate, and process different types of waste, including those from homes and institutions.

Although traditional waste management methods rely on manual sorting at Material Recovery Facilities (MRFs), this process is still not very effective and can lead to contamination.
Also, many existing educational programs aiming to reduce waste often fail to bring about meaningful changes in people's behavior within the waste management system. To bridge this gap between knowing what should be done and actually doing it, this project introduces a new, user-friendly, web-based application that uses gamification techniques to encourage behavioral change.

By including gamified elements such as leaderboards, reward points, and real-time feedback, the application aims to help users correctly identify and dispose of waste.
This approach aligns with the Sustainable Development Goals (SDGs), specifically Goal 11 (Sustainable Cities and Communities) and Goal 12 (Responsible Consumption and Production), as shown in recent studies. The platform's innovative design allows individuals, families, and schools to actively participate in creating a sustainable economy where materials are continually reused, reducing the need for new resources and decreasing environmental waste.

Ultimately, this project is not just about managing waste as a simple task, but about creating a fun, competitive, and educational experience that helps change people's mindset toward sustainable development.

CHAPTER 2

RELATED STUDY

Researchers are continuing to search for ways that technology, such as smartphone apps that use gamification or artificial intelligence, can encourage greater involvement in tackling problems like urbanization and waste generation caused by increasing city populations.
Previous methods of waste sorting, like physical sorting, could not use digital interaction or effectively change people's psychological behaviors.

Researchers like Kasurinen and Knutas (2018) believe that gamification can boost student motivation and increase environmental responsibility by incorporating game features such as points, badges, levels, and rewards.
This idea—that users who receive virtual points for environmentally friendly actions will continue to do so—has been supported by several studies (e.g., Gunter et al. 2007, Shen et al. 2010). However, it's important to note that for users to consistently recycle and dispose of waste correctly, they need to remain involved and feel a real sense of achievement from their actions (e.g., Keller et al. 2005). Gamification alone cannot provide long-term motivation, which means the long-term benefits of using gamification to promote eco-friendly behavior may be limited.

Supporting this view, behavioral science shows that gamification can be an effective way to develop a digital waste management system.
As noted by Sheeran et al. (2008), the Theory of Planned Behaviour (TPB) suggests that a person’s willingness to recycle depends on their beliefs about recycling, the expectations of others regarding recycling, and their ability to perform it. Gamified apps help by offering users instant feedback, tracking progress, and providing social recognition—factors that support these beliefs (Gunter et al. 2007; Keller et al. 2005). Examples of gamification in waste disposal include Junker (Junker 2011) and GreenApes (GreenApes 2011), which use barcode scanning, progress dashboards, and community challenges to help users make better decisions about waste.

While technological progress has greatly improved the ability to collect and process recyclables, the continued use of traditional systems presents new challenges.
Recent research shows that CNNs can accurately classify waste items using images. Once trained on enough data, these CNN systems can correctly identify recyclable, biodegradable, plastic, metal, and hazardous items. These advanced technologies have reduced human errors in classification, minimized contamination of recyclables, and made waste processing more efficient compared to manual systems.

Support for using smart, IoT-enabled waste bins is also growing, especially with the addition of sensors.
These smart bins can detect when they are full and track what items are placed in them. When a bin is nearly full, it can automatically notify waste collection services, which helps optimize collection routes, reduce fuel costs, and create cleaner public spaces. Combining technology with gamification through an AI-based waste sorting system gives users an easy and accurate way to separate their waste from the correct bin. Additionally, research shows that through community support, technological innovations can help achieve long-term sustainability by promoting and assessing environmentally-friendly practices.

Social engagement via digital platforms is also necessary to achieve environmentally sustainable products and strengthen this sense of collective responsibility. The integration of behavioural science, social influence, and artificial intelligence creates an ecosystem to support sustainable waste management.Numerous recent studies illustrate the evolution of digital technology within the waste management industry—the movement from an objective, task-oriented activity to an engaging and intuitive aspect of our daily lives. Previous waste management methods often involved many manual labour processes (sorting, awareness campaigns) and relied heavily on the community's lack of knowledge about how to dispose of their recycled products correctly. Although these past strategies may have raised people's overall awareness of recycling, they did not typically engage participants at the time they were disposing of materials.

In response to Digital Sustainability Tools, research has demonstrated that users develop more consistently correct sorting behaviour quicker when they receive Real Time Feedback—for example, to see if a given item is recyclable. Digital Sustainability Tools may also provide instant feedback and enhance the user experience because they remove confusion about how to dispose of items properly, so this approach provides far better results for both the user community and the environment. Digital Technology such as Artificial Intelligence-enabled cameras helps to eliminate the challenge of identifying what to do with complex or misleading packaging materials. Digital Technology also facilitates a smooth user experience and allows households and students to engage in responsible practices easily.

Research in behavioural science has demonstrated an additional and very important factor we need to consider: people respond much better to continuous forms of motivation than they do to

simply one-time perceptions. Gamification appeals to this need through the introduction of reward systems, badges, challenges and a sense of accomplishment. A number of recent studies have shown that the presence of these types of features greatly enhances the level of participation, particularly among adolescents and young adults.

Once people see others competing and having fun, they are much more likely to continue participating even after the "honeymoon" period of gamification has passed. There is also considerable and growing evidence to suggest that social influence is a major contributor to people's motivation to take environmental action. When individuals can compare their progress to others, such as friends and family members or school groups, this motivates them to increase their level of participation. The gamified applications that have included features such as leader boards or community challenges have achieved higher rates of user participation than those that provide only educational content. This social reinforcement is a powerful tool to assist and motivate entire groups of individuals rather than just individual users.

Cloud computing allows systems to handle vast amounts of information, store it, process it and analyze it. By tracking waste generation, user behavior over time, and the rate of error over time, management can make better informed decisions based on the trends presented by this data. This model of making decisions based on data analysis will promote the long-term sustainability of urban waste by allowing policy makers to determine what gaps exist, identify what the immediate and future needs will be, and create better urban waste policies.

The involvement of the community has become one of the major impacts on waste management research. People who participate in collective activities, such as cleaning up areas, improving their recycling rate, and receiving recognition for their contribution to the activities, tend to continue to be involved in those activities. Utilizing digital platforms to support local achievements, cleanup efforts, and event-based competitions builds a stronger relationship between people and their environment.

Emerging trends in sustainability technologies suggest that the future of waste management will become even more personalized and AI-augmented. Researchers are exploring advanced features like AR-based sorting instructions, behaviour prediction models, and adaptive learning modules that change based on user performance. These innovations aim to make waste disposal not only accurate but also enjoyable, reducing the burden on municipal systems.

In summary, research at the global level demonstrates that it is important to incorporate gamification, artificial intelligence-based recognition, and smart monitoring technologies into the existing framework for managing waste. These solutions will bridge the gap between how people act and how waste is managed through current approaches. They also provide a more engaging learning environment for individuals and support their achievement of broader environmental objectives. Building cleaner, smarter, and more sustainable communities requires the adoption of a human-focused and technological approach.

CHAPTER 3

FEASIBILITY ANALYSIS

Technical Feasibility:

The technology used for the proposed gamified waste management and waste identification system is readily available today through modern and accessible technologies. Image recognition using artificial intelligence models has progressed so far that training them no longer requires high-end systems. Pre-trained convolutional neural networks can now be used to accurately classify types of waste with little to no computational cost. All the development technologies (Python, TensorFlow or PyTorch for AI, React for the interface and Node.js for the backend service) are also very popular with lots of developers creating applications using them

every day. The IoT elements, for example ultrasonic sensors, weight sensors, and RFID modules, are inexpensive and easy to install. The aforementioned also provide evidence that the project is technically viable as it makes use of the existing technology hence has a very low technical barrier to entry.

Operational Feasibility:

The system has been developed in a manner to ensure that it fits seamlessly into the existing daily routines of students, families, and community members, so as not to require a major change in routine for the participant. The only tool the user will need in order to use the AI scanner or engage with the gamified programs will be their smartphone; therefore, the system will require very little complexity. In addition, the administrative staff and teachers will have an online coaching dashboard that will allow them to monitor user points, the accuracy of waste classification, and the level of user participation at any given time.Since the system does not require any special skills to operate, nor does it have any complex equipment, this allows institutions to maintain a low operational workload. As a result, the average user can easily adopt and continue to utilize the system.

Economic Feasibility:

The economic feasibility of the system is very positive because of the absence of major costs. The primary cost of the system will be in software development; however, due to the digital and automated nature of the system, the ongoing costs of maintenance are minimal. The AI systems that will be on the platform can be trained using publicly available datasets, which means that there is no need for the purchase of very expensive proprietary data. The IoT smart bin devices, while being optional, have become significantly more affordable now that there have been many advances in sensor technology. In addition to providing long-term cost savings compared to traditional awareness programs that required numerous prints of posters, or manual monitoring of communities, and the expenditure of human labour to conduct workshops, using the digital platform will provide significant savings to local governments and organizations. Additionally, communities that are successful at utilizing proper waste segregation experience economic benefits such as lower landfill disposal fees and, therefore, a significant increase in the quality of materials that can be recycled.

Behavioural Feasibility:

The system design is consistent with numerous psychological models that explain human behaviour and provide insight into how to change behaviour. Gamification techniques (points, badges, levels, and leaderboards) create consistent incentives for users to engage with the platform. As well as there being many studies to show that the effects of providing a user with

immediate feedback create a stronger foundation for both learning and forming habits of disposing of waste in an appropriate manner.

Social Feasibility:

The feasibility study assessment of the system's social aspects indicates that there are a number of positive factors in its favour, principally that waste management is becoming increasingly important in the eyes of households and schools and that there is also a growing awareness about how people should recycle within their communities. In addition to this, the system fosters the co-operation and collaboration of many users through group challenges and competitions between schools and communities. As community members can see their part in the overall progress of the community, their motivation to continue to use the system will be enhanced.

Environmental Feasibility:

Teachers, Parents, and Local Authorities will have access to a structured format for helping children learn about improving their environment, while children are encouraged to participate in a healthy and supportive environment. As there is no social pressure associated with using the system, it would appear that user acceptance should be relatively high across all age groups.

There are several very positive environmental benefit factors associated with this system which make it a very viable and sustainable solution to the many issues which affect our planet today. By improving waste identification, the contamination levels of recyclable materials and compostable waste will be significantly lessened as they will be sorted into their correct categories, improving the potential for recycling. It will also significantly reduce the number of tons of waste going to landfill. Smart IoT enabled bins will allow for quicker collection of overfilled bins. By allowing users of the platform to learn how to improve waste categorization at home, the platform can be used as a direct support for all national waste improvement and Sustainable Development Goals.

Legal and Ethical Feasibility:

The Project supports the existing guidelines for Waste Management in India and the Solid Waste Management Rules initiative, both of which encourage source segregation and education/awareness. There are no legal impediments to the project that will prevent an Educational Tool or Gamification Method from developing and implementing AI and Machine Learning capabilities as part of the program. The application only collects general user data and

therefore can meet ethical requirements when proper Transparency and Privacy Best Practices are followed. There are no Academic or Digital Governance limitations that would prevent a School or Institution from adopting the Project.As such, since the Project supports both Education and Sustainable Development Goals, it is both ethically responsible and legally compliant.

Scalability Feasibility:

The Project will be highly scalable due to its modular design. Schools will be able to implement the Project by using the app for Waste Identification only and over time, driving any significant interface performance enhancements by implementing IoT Smart Waste Bins into the Platform. The Cloud infrastructure will allow the Project to support increasing user volume as more communities come on board. Additionally, new features will be able to be added to the Project as user feedback is captured (i.e. location-based rewards, AR-based sorting guidance, multiple languages, and integration with Municipal Waste Dashboards), without requiring an entire redesign of the system. These attributes enable the Project to grow from a small, pilot Program/Project to a large and impactful Environmental Tool.

User Acceptability Feasibility

The acceptance of users is key to whether users will enthusiastically adopt the system; gamification enhances the potential of making that happen by attracting and keeping users' interest. Generally, younger users prefer to use game-like and interactive methods of learning compared to reading lengthy instruction manuals and attending lectures. The first steps the system takes to promote user familiarity to the point of building user confidence reside with identifying waste and reducing any possible confusion about the definitions and types of waste. The education benefits to teachers and parents of the system are demonstrated through the fact that students can earn rewards for their successes while using the system. Therefore, the

combination of the value of education through learning and the allure of earning rewards will naturally encourage users to use the system.

Long-Term Feasibility:

The ability for individuals using this platform to successfully establish themselves as environmentally conscious members of society is what will drive future success with respect to the continued growth of the platform. This is accomplished through the development of habits based upon the use of the gamification aspect of the system, which provides users with an ongoing experience of continuing to develop good eco-friendly behaviours. Also, AI technologies continue to be developed at a very rapid pace; therefore, the platform will have access to, and will utilize, new and improved AI-generated models in the not- too-distant future. Additionally, an increase in the number of individuals who participate within the eco-community will help strengthen the community as individuals become members. Furthermore, Integration of the platform into educational programs (i.e., eco-clubs) and neighbourhood programs (i.e., community initiatives) will also contribute to the continued growth and long-term sustainability of the platform.

CHAPTER 4

REQUIREMENT ANALYSIS

The increasing use of gameplay elements to enhance the way in which individuals learn, engage with other people, and make choices has dramatically altered the way in which individuals interact with technology. As more than two billion people engage with some form of entertainment-based gaming around the world, it is not surprising that researchers and educators

are now focusing on the positive impact of utilizing gaming as a means of: (1) motivating individuals to be active participants in their own learning; (2) enhancing the ability of individuals to transfer knowledge and skill into real-world applications; and (3) motivating individuals to adopt sustainable behaviours through the use of Game-Based Learning (GBL) and Features of GBL. This chapter will describe a detailed requirement analysis of a proposed system that utilizes GBL and components of GBL to assist learners with increasing their motivation to learn and achieve improved learning outcomes while simultaneously fostering creativity and promoting a sustained effort to practice sustainable behaviours. This requirement analysis will identify both functional and non-functional requirements for developing an immersive educational experience that can be applied to a variety of learner populations.

There exists a growing body of research that demonstrates the advantages and disadvantages of using GBL and/or Components of GBL as a means of increasing learner motivation and enhancing engagement, problem-solving, and performance. Although GBL and Components of GBL have strong evidence supporting their effectiveness, research also indicates that poorly designed GBL or Components of GBL can lead to reduced learner satisfaction and decreased performance results. Therefore, developing a comprehensive requirement analysis of GBL and Components of GBL will provide a framework for creating a balanced, meaningful and educationally sound solution.

System Overview

The system has been created to provide educational content while simultaneously implementing behaviour change interventions through a digital environment that is interactive and engaging. There are two main ways the system will do this, via gamified learning which embeds the instructional content into a full-blown gaming experience and gamification of regular tasks through the addition of game-like elements. Using this gamification approach, it is possible to

create a sense of continued user interest and motivation to stay engaged with the platform for an extended period of time due to the inclusion of motivational elements such as points, levels, badges, leaderboards, missions, and challenges. Additionally, the solution is appropriate for education, sustainability training, behavioural training, and development of skills that emphasise creativity, communication, teamwork, and critical thinking. To maintain learner engagement and improve learner outcomes through reinforcement, the system contains various feedback loops, dynamic difficulty adjustments, and other interactive features that allow for real-time adjustment to user performance across all activity levels.

Stakeholder Analysis

All stakeholders must be considered to make sure the system meets their individual needs. Primary users are learners - students, general users, and employees. They need access to an easy-to-use, enjoyable, and intuitive interface in order to navigate through learning processes and have activities that can easily be completed with positive reinforcement through rewards.. Lastly, Researchers/evaluators require access to anonymized data sets for studying behavioral trends, measuring performance patterns, and analyzing how the system affects Sustainable Behavior and Learning Outcomes

Functional Requirements

To achieve its objectives, the proposed system should provide a secure mechanism for registering users, authenticating users, and managing user profiles. Progression and accomplishments of users, task completions, and each user's history of interactions with the system should be documented in a reliable storage device(s) for purposes of ongoing tracking and evaluation.

The platform will provide a selection of interactive/educational modules. The types of modules may include written materials, images, quizzes, puzzles, simulations, as well as multi-stage learning challenges. Feedback from the platform will include hints, explanations as well as adaptive progression. Gamification elements (e.g., missions, levels, unlockable stages, mini-games, animations, audio feedback) will immerse users in the learning experience and enhance engagement through motivation to return and continue using the application by virtue of points, badges, certification levels, leaderboards, streaks, boosters, and other personalized recommendations. Monitoring Tools: The system incorporates monitoring tools that assess user behaviour, user actions, and user performance by how a user interacts with the task. For example, tasks that are centred on Sustainability show how accurately a user sorts their waste, what choices a user makes to conserve energy, etc. On the other hand, tasks that measure

Creative Thinking, the system evaluates the degree of originality, approach to solving problems, and teamwork (collaborative effort). The dashboards that educators access will provide them with a detailed breakdown of a student or user's performance trends and other engagement statistics, strengths, weaknesses, completion rates, and creativity/collaboration/communication indicators; all of which can be exported to meet the academic or institutional needs of the educators and schools for monitoring progress. In addition, by using the systems social and collaborative tools, educators can assign their classes group activities, projects, team challenges, peer-to-peer interactions, and shared accomplishments, as well as leaderboards for team-based competition. This will ultimately provide educators with a way of encouraging class collaboration to maximise their ability to work together on solving problems and improving their problem-solving skills. Administrators have access to update content, monitor security logs, manage server performance, and enforce data privacy.

The system must operate in an effective manner functionally, meaning it must load fairly quickly (less than 3 seconds from the time a user clicks a link until he/she sees content) and should remain responsive regardless of the volume of traffic. To protect the ultimate confidentiality of both user information and user activity, security measures such as encryptions, secure login methods, and other forms of authentication (including Defense in Depth) must be implemented to prevent unauthorized access to user information. The user interface must also be designed to appeal visually to users of all ages and be simple to navigate for all users. Tutorials and tooltips should be used to support the onboarding of new users so they can learn how to use the platform. Reliability will be confirmed with consistent uptime for the system, scheduled backups of data, and the ability to recover and save a user's progress in the event the system fails or otherwise interrupts service to a user and disrupts that user's progress.

Non functional Requirements

Numerous research studies have found that using gamification to provide motivational, enjoyable, efficient & long-term engagement has proven effective. In fact, those who participate in gamified systems have improved environmental awareness and exhibited continuous positive behaviours for an extended period of time. In addition, gamified systems educate users about principles of waste management and allow users to integrate that knowledge into day to day life resulting in developing sustainable habits through practising.

In addition to changing individual behaviour, the gamification model addresses many of the UN's Sustainable Development Goals (SDG). Those goals include promoting responsible

consumption, protecting the environment, combating climate change, and providing quality education. The intended audience of this program is young people; therefore, the system aims to support youth development at a pivotal point in their lives by providing them with the knowledge, skills and motivation needed to contribute to a greener and more sustainable tomorrow.

CHAPTER 5

PROTOTYPE

The gamified leaderboard application was professionally developed to be both motivating and fun for users through competition based on data. The gamified nature of the application is built on core features that include points, levels, badges and rankings to keep users engaged and provide a sense of accomplishment. The UI was developed with a user-centric experience, providing personalized statistics of the user's performance while allowing users to view who the top performers are, promoting friendly competition within the eco-warrior community.

The gamified leaderboard application is built with modern frontend development techniques with the use of React to provide a clean, well-organized structure in the code base and an aesthetically-pleasing look and feel through Tailwind CSS. Additionally, this application takes advantage of the state and props features of React to build dynamic content updates on the leaderboard so that users are always viewing an up-to-date leaderboard.

The top three podium positions are displayed with unique visual styles and icons, providing users with an easy way to identify where they rank compared to others, while the overall leaderboard is presented in a scroll-down ranking format. Users can easily see how they are doing versus everyone else since all users are shown on the leaderboard and users will see their statistics.

```
 import { useState } from 'react';
import { LandingPage } from './components/LandingPage';
import { Onboarding } from './components/Onboarding';
import { Dashboard } from './components/Dashboard';
import { ScanBarcode } from './components/ScanBarcode';
import { PhotoClassifier } from './components/PhotoClassifier';
import { Leaderboard } from './components/Leaderboard';
import { Rewards } from './components/Rewards';
import { Quiz } from './components/Quiz';
import { Profile } from './components/Profile';
import { AdminDashboard } from './components/AdminDashboard';
import { SplashScreen } from './components/SplashScreen';
import { MobileNavBar } from './components/MobileNavBar';
import { Login } from './components/Login';

export type UserProfile = {
  name: string;
  username: string;
  interests: string[];
  goals: string[];
```

```
  points: number;
  level: number;
  badges: string[];
  wasteLog: WasteEntry[];
};

export type WasteEntry = {
  id: string;
  type: 'plastic' | 'metal' | 'compost' | 'general' | 'glass' | 'paper';
  item: string;
  date: Date;
  points: number;
  correct: boolean;
};

export type Screen = 'landing' | 'onboarding' | 'dashboard' | 'scan' | 'photo' | 'leaderboard' | 'rewards'
| 'quiz' | 'profile' | 'admin';

export default function App() {
  const [showSplash, setShowSplash] = useState(true);
  const [currentScreen, setCurrentScreen] = useState<Screen>('landing');
  const [userProfile, setUserProfile] = useState<UserProfile | null>(null);
```

```
  const [isAdmin, setIsAdmin] = useState(false);


  const handleStartOnboarding = () => {
    setCurrentScreen('onboarding');
  };


  const handleOnboardingComplete = (profile: Omit<UserProfile, 'points' | 'level' | 'badges' |
'wasteLog'>) => {
    setUserProfile({
      ...profile,
      points: 0,
      level: 1,
      badges: ['newcomer'],
      wasteLog: []
    });
    setCurrentScreen('dashboard');
  };


  const handleItemScanned = (entry: Omit<WasteEntry, 'id' | 'date'>) => {
    if (!userProfile) return;


    const newEntry: WasteEntry = {
```

```
  ...entry,
  id: Date.now().toString(),
  date: new Date()
};

const newPoints = userProfile.points + entry.points;
const newLevel = Math.floor(newPoints / 100) + 1;
const newBadges = [...userProfile.badges];

// Award badges based on milestones
if (newPoints >= 100 && !newBadges.includes('eco-warrior')) {
  newBadges.push('eco-warrior');
}
if (newPoints >= 500 && !newBadges.includes('planet-protector')) {
  newBadges.push('planet-protector');
}
if (userProfile.wasteLog.length + 1 >= 10 && !newBadges.includes('sorting-master')) {
  newBadges.push('sorting-master');
}

setUserProfile({
```

```
    ...userProfile,
    points: newPoints,
    level: newLevel,
    badges: newBadges,
    wasteLog: [newEntry, ...userProfile.wasteLog]
  });
};

const handleRewardRedeemed = (points: number) => {
  if (!userProfile) return;
  setUserProfile({
    ...userProfile,
    points: userProfile.points - points
  });
};

const handleQuizComplete = (points: number) => {
  if (!userProfile) return;
  setUserProfile({
    ...userProfile,
    points: userProfile.points + points
```

```
  });
};

const handleNavigate = (screen: Screen) => {
  setCurrentScreen(screen);
};

// Screens that should show mobile nav
const screensWithNav: Screen[] = ['dashboard', 'leaderboard', 'rewards', 'profile', 'admin'];
const showMobileNav = userProfile && screensWithNav.includes(currentScreen);

if (showSplash) {
  return <SplashScreen onComplete={() => setShowSplash(false)} />;
}

return (
  <div className="min-h-screen bg-gray-50">
    {/* Mobile Status Bar Spacer */}
    <div className="h-safe-top bg-transparent"></div>

    {currentScreen === 'landing' && (
```

```
  <LandingPage onGetStarted={() => handleNavigate('onboarding')} />
)}

{currentScreen === 'onboarding' && (
  <Onboarding onComplete={handleOnboardingComplete} />
)}

{currentScreen === 'dashboard' && userProfile && (
  <Dashboard
    userProfile={userProfile}
    onNavigate={handleNavigate}
    isAdmin={isAdmin}
  />
)}

{currentScreen === 'scan' && userProfile && (
  <ScanBarcode
    onBack={() => handleNavigate('dashboard')}
    onItemScanned={handleItemScanned}
  />
)}
```

```
{currentScreen === 'photo' && userProfile && (
  <PhotoClassifier
    onBack={() => handleNavigate('dashboard')}
    onItemClassified={handleItemScanned}
  />
)}

{currentScreen === 'leaderboard' && userProfile && (
  <Leaderboard
    currentUser={userProfile}
    onBack={() => handleNavigate('dashboard')}
  />
)}

{currentScreen === 'rewards' && userProfile && (
  <Rewards
    userProfile={userProfile}
    onBack={() => handleNavigate('dashboard')}
    onRewardRedeemed={handleRewardRedeemed}
  />
```

```
)}

{currentScreen === 'quiz' && userProfile && (
  <Quiz
    onBack={() => handleNavigate('dashboard')}
    onQuizComplete={handleQuizComplete}
  />
)}

{currentScreen === 'profile' && userProfile && (
  <Profile
    userProfile={userProfile}
    onBack={() => handleNavigate('dashboard')}
    onNavigate={handleNavigate}
  />
)}

{currentScreen === 'admin' && isAdmin && (
  <AdminDashboard
    onBack={() => handleNavigate('dashboard')}
  />
```

```
      )}

      {/* Mobile Navigation Bar */}
      {showMobileNav && (
        <MobileNavBar
          currentScreen={currentScreen}
          onNavigate={handleNavigate}
          isAdmin={isAdmin}
        />
      )}
    </div>
  );
}
```

CHAPTER 6
TIME LINE ACTIVITY

S.NO	Activity list	Date/s of activity completed	No. of hours spent
1.	Reviewing Waste Management Concepts	05.08.2025	1
2.	Introduction Draft preparation	12.08.2025	1
3.	Related Study Collection	19.08.2025	2
4.	Requirement Analysis Work	09.09.2025	2
5.	Design /Prototype Drafting	23.09.2025	3
6.	System WorkFlow Structuring	21.10.2025	2
7.	Feasibility Study Preparation	11.11.2025	3
8.	Technology Description Writing	18.11.2025	2
9.	Compiled references in appropriate format	25.11.2025	1
10.	Document Finalization	02.12.2025	1

CHAPTER 7

CONCLUSION

This report presents an extensive examination of how gamification strategies can be leveraged to improve waste conversion practices by transforming routine waste-management tasks into enjoyable, meaningful, and interactive experiences. The main concept here is to create a system that combines environmental awareness with aspects of play, motivation, and reward. By using game elements such as achievement badges, points for progress, challenges, and leaderboards, this system intends to generate interest in waste sorting/recycling/reporting and to keep students, families, and members of the community involved and invested in the system at all times. Through the use of game elements, waste sorting/recycling/reporting will be an activity that is rewarding rather than a burden.

To help with this approach, the report describes how some of the latest advancements in digital technology (e.g., data analytics, AI, cloud computing, IoT devices, image recognition, etc.) allow for the creation of user-friendly platforms that will be able to support individuals in determining what type of waste they are throwing away; track their waste disposal habits; and receive timely feedback on their waste disposal behaviors. As a result of their scalability and adaptability, these platforms can provide services across educational institutions, in the home, and in the broader community; therefore, they will be available to users on a daily basis.

Through the literature review and case studies cited, it can be seen that gamification results in much greater engagement with waste-management activities and provides significant improvements in user performance, environmental awareness, and long-term commitment to environmental behavior. The use of gamification provides users with the opportunity to progress from having a passive understanding of waste management to actively implementing waste-management practices. This is a considerably different shift in the learning process.

CHAPTER 8

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